Maths Week 4: dividing fractions (written)

Ember · Non-calculator · about 15 minutes

Date: _____

Skill: written method and checking. Write every step. Under each answer, write a check: multiply your answer by the number you divided by. You should get back to where you started.

[2 marks]

1. Work out $\frac{7}{8} \div \frac{1}{4}$

Give your answer as a mixed number.

Working

Answer _____

Check: _____

[3 marks]

2. Work out $2\frac{2}{5} \div 1\frac{1}{5}$

Working

Answer _____

Check: _____

[3 marks]

3. A ribbon is $4\frac{1}{2}$ m long. It is cut into pieces that are each $\frac{3}{4}$ m long. How many pieces are there?

Working

Answer _____ pieces

Check: _____

[2 marks]

4. Sam says, "When you divide, the answer is always smaller than the number you started with." Use $\frac{1}{2} \div \frac{1}{4}$ to show that Sam is wrong. Explain your answer.

Self-check (5 min)

□ I changed mixed numbers to improper fractions first.

□ I flipped the second fraction, not the first.

□ I wrote a check under questions 1 to 3, and it worked.

□ Question 4: I worked out the answer **and** explained why it is bigger.

□ Reflection: which step do I most often forget?

Answers

- **1.** $\frac{7}{8} \times \frac{4}{1} = \frac{28}{8} = \frac{7}{2} = 3\frac{1}{2}$ **(M1 A1)**. Check: $3\frac{1}{2} \times \frac{1}{4} = \frac{7}{8}$ ✓
- **2.** $\frac{12}{5} \div \frac{6}{5} = \frac{12}{5} \times \frac{5}{6} = \frac{60}{30} = 2$ **(M1 M1 A1)**. Check: $2 \times 1\frac{1}{5} = 2\frac{2}{5}$ ✓
- **3.** $\frac{9}{2} \div \frac{3}{4} = \frac{9}{2} \times \frac{4}{3} = \frac{36}{6} = 6$ pieces **(M1 M1 A1)**. Check: $6 \times \frac{3}{4} = \frac{18}{4} = 4\frac{1}{2}$ ✓
- **4.** $\frac{1}{2} \div \frac{1}{4} = \frac{1}{2} \times 4 = 2$, which is bigger than $\frac{1}{2}$ **(1)**. Because $\frac{1}{4}$ is less than 1, more than one quarter fits into a half: two quarters fit, so the answer is bigger **(1)**.

Score: _____ / 10

Questions written for Ember in AQA style. Week 4 screen questions include AQA Mathematics 8300/1H November 2020 Q4 © AQA, used for private practice.